SPE-230141-MS

Saudi Aramco's Largest and Heaviest Offshore Jackets – Marjan Gas Oil Separation Plant #4

A. Albahrani, A. Ruhaili, H. Shaiban, and A. AlShaye, Saudi Aramco

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

The Marjan Increment Program's development of the Marjan Gas Oil Separation Plant No. 4 (MGOSP-4) is set to become one of the world's largest offshore complexes. The development includes the fabrication, offshore installation, and commencement of six primary jackets that constitute the base of the new GOSP. This paper outlines the challenges and the mitigation undertaken by the project team to overcome and achieve the successful fabrication and installation completion.

The fabrication challenges include but not limited to, large dimension and heavy weight of the structures, fabrication yard space constraint and challenging completion schedule to meet the installation target date. The installation challenges include the weight and dimension and precision requirement to ensure the alignment of the topsides and bridges. To overcome these challenges the project implemented several techniques and technologies that are detailed in the paper. These include, the utilization of material handling tracker and QR based weld consumables management system, automatic welding Tanique's, zero gap welding approach, augmented reality training for working at height, and jacket installation simulation.

As a result of the utilizing the abovementioned Tanique's, the project met the fabrication and installation completion target date with zero lost time injury. The QR code technology facilitated real-time tracking of welding consumables through the scanning of unique QR codes, leading to enhanced control, improved welder productivity, efficiency, and automation. This resulted in weld automation boost from 55% to 75%, reporting efforts were streamlined, and welding productivity soared by approximately 30%. Furthermore, approximately 300 metric tons of welding consumables were issued through the QR coding system, providing enhanced accountability, and real-time monitoring of stock status. Moreover, the welding process was further optimized through the implementation of Zero Root Gap Welding, where the Groove design is made such that no root gap with 2-4 mm root face. The welding process demonstrated improved speed and efficiency compared to other welding methods. This approach facilitated the even dispersion of heat and effectively minimized the chances of distortion or warping. This enhancement translated to 30% increase in productivity, evident across more than 2800 joints that were welded using a zero-root gap technique. Lastly, the simulation of the jacket installation supported and safe and efficient installation of the structures at offshore.

This paper will provide details of the utilized technologies and techniques that can serve as a reference for future fabrication and installation projects for Mega Offshore structures.

Key words: Offshore Construction, Fabrication Safety, Augmented Reality

Introduction

Part of Saudi Aramco's Marjan Increment Mega Program is the Marjan Offshore GOSP-4 Development Project (MGOSP-4). MGOSP-4 scope includes the engineering, procurement, fabrication, installation and hookup of Marjan Gas Oil Separation Plant #4 (MRJN GOSP-4) facilities consisting of 6 major topside platform decks and jackets, interconnecting bridges, flare platform, trunkline, submarine composite cables in Marjan Field along with associated onshore infrastructure at onshore Tanajib Plant. The GOSP-4 is located at Marjan Oil Field in the Arabian Gulf at water depths reaching to 52 meters.

As the construction phase of this project commenced the project strategically distributed the fabrication locations into multiple fabrication yards across the globe. The Jackets which are the submerged steel structures used as a foundation for the offshore platforms in shallow water depths, has been subcontracted by the main Contractor "McDermott Middle East" to be the subcontractor "Larsen and Toubro" in their fabrication yard at Sohar, Oman. The yard has been selected for its large open space that can accommodate the ground fabrication of the all the 6 mega jackets in parallel.

A significant achievement has been reached in Marjan Increment Program specifically in the development of Marjan Gas Oil Separation Plant #4 (MRJN GOSP-4), poised amongst the world's biggest offshore complexes. This achievement entails the successful fabrication completion and Offshore installation for the six (6) primary jackets that constitute the base of the new GOSP. These jackets hold the distinction of being the largest and heaviest offshore Jackets in the history of Aramco.

The journey of these remarkable Jackets commenced in December 2021, when the Marjan Increment Projects Department (MIPD) celebrated the steel cutting ceremony in Sohar, Oman. This marked the beginning of fabrication activities of the six (6) Jackets for the Marjan Offshore GOSP-4's scope. These Jackets are the largest ever built and installed Jackets in Saudi Aramco's offshore fields, collectively weighing a substantial 33,000 metric tons. Their individual weights range from 3,800 to 6,000 metric tons per jacket, three times the average weight of conventional tie-in jackets. With dimensions reaching up to 88 × 50 × 57 meters, these eight (8) and ten (10) legs floatover jackets were vertically fabricated simultaneously in the yard. Please refer to the below table for the details of grid spacing for the Jackets.

Sl No	Platform Name	Jacket Grid Spacing in Transverse Direction (meter)	Topside Overall width in Transverse Direction (meter)
1	TIE IN Platform	50	62
2	EDP Platform	50	62
3	PRODUCITON Platform	50	62
4	GAS COMPRESSION Platform	50	62
5	AUXILIARY Platform	50	62
6	ACCOMMODATION PLATFORM	48	53
Sl No	Platform Name	Jacket Grid Spacing in Longitudinal Direction (meter)	Topside Overall width in Longitudinal Direction (meter)
1	TIE IN Platform	18 – 18 - 18	74.0
2	EDP Platform	15 – 20 - 15	66.0
3	PRODUCITON Platform	22 – 22 – 22 - 22	96.60
4	GAS COMPRESSION Platform	25 – 25 - 27	86.0
5	AUXILIARY Platform	18 – 18 - 18	69.0
6	ACCOMMODATION PLATFORM	22 – 22 - 22	66.0

Figure 1—Marjan GOSP-4 Jackets Grid Spacing

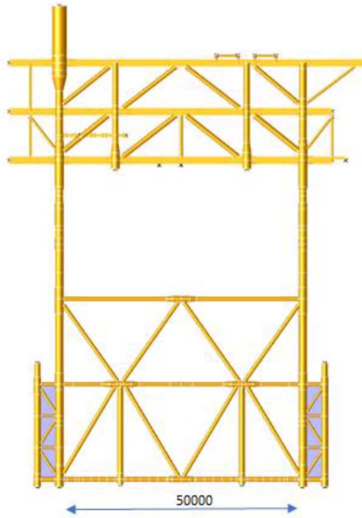

Figure 2—Transverse Direction Jacket Grid Spacing

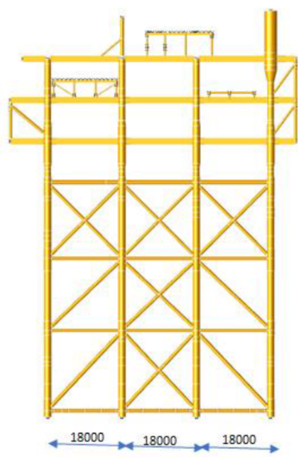

Figure 3—Longitudinal Direction Jacket Grid Spacing

Fabrication Journey from challenges to on-ground achievements

The substantial size and scale of the jackets presented a materials management challenge, spanning from procurement to production. The project commenced during the complex backdrop of the COVID-19 pandemic, marked by a pronounced scarcity of supplied tubular materials required for initiating the jackets' fabrication. Nonetheless, leveraging a series of comprehensive constructability reviews, the team adeptly overcame this hurdle by identifying alternative work approaches that utilizes the available materials. Modularization played a pivotal role in enabling concurrent work fronts, culminating in the timely achievement of the sail-away date for the Jackets. Upon the material's arrival on-site, the team achieved a remarkable fabrication progress of 54.11% within a span of just five months.

Another challenge lay in the sheer volume of welding activities necessary for the jackets' fabrication. To overcome this challenge, the project established a dedicated welding engineering department on-site, comprising a team of welding subject matter experts and skilled personnel. This department effectively controlled all site welding activities and maximized the use of automatic welding techniques, resulting in a reduction of total weld repair percentages. Noteworthy outcomes include the completion of panel (sub) welding within six shifts, half-cut tubular welding in four days, and an increase in Shielded Metal Arc Welding (SMAW) (6GR) welders' performance to four joints in place of the conventional two joints on 24"-36" diameter tubulars.

Moreover, the welding process was further optimized through the implementation of Zero Root Gap Welding, where the Groove design is made such that no root gap with 2-4 mm root face. The welding process demonstrated improved speed and efficiency compared to other welding methods. This approach facilitated the even dispersion of heat and effectively minimized the chances of distortion or warping. By eliminating the need for manual SMAW in the Root and Hot passes, in addition to streamlining joint preparation, the overall quality of welded joints was elevated. This enhancement translated to an impressive 30% increase in productivity, evident across more than 2800 joints that were welded using a zero-root gap technique. As a result below key achievements were possible:

- Panel (sub) welding completed in 6 shifts
- Half-cut tubular welding completed in 4 days.
- Welders' performance doubled on 24"-36" dia tubulars.
- Leg welding completed in record time;2 weeks.
- 600+ joints NDT tested in weekly basis

Material Handling Tracker and QR Based Weld Consumables Management System

The project team embraced advanced technologies on the fabrication journey, including the utilization of Material Handling Equipment Tracker and QR based Weld Consumables Management Systems. Notably, the QR coding system emerged as a digital platform for efficient control of welding consumables, seamlessly integrated with a Data Base Management System. This technology facilitated real-time tracking of welding consumables through the scanning of unique QR codes, leading to enhanced control, improved welder productivity, efficiency, and automation. The results of these technological implementations were profound, as welder control was centralized, weld automation was boosted from 55% to 75%, reporting efforts were streamlined, and welding productivity soared by approximately 30%. Furthermore, approximately 300 metric tons of welding consumables were issued through the QR coding system, providing enhanced accountability, and real-time monitoring of stock status. The technologies implementation within the Marjan Increment Program exemplify Saudi Aramco's commitment to innovation and its continuous pursuit of excellence.

Augmented Reality Training for Working at Height

In the construction industry, working at height incidents are a significant contributor to fatalities and injuries. Falls from heights consistently rank among the leading causes of construction-related fatalities worldwide. In response, the fabrication strategy for the Jackets was meticulously developed with a focus on minimizing elevated work with more than 85% of the work done on ground. However, the remaining 15% works at height cannot be eliminated and for that portion the project conducted multiple focused safety training for working at height utilizing augmented reality technology. This technology offered a dynamic alternative to conventional training that provides risk-free environments and scenarios allowing employees to sharpen the skills required for the job and increase the safety awareness amongst the team members. This approach resulted in achieving more than 9 million safe working manhours without Loss Time Injury.

The jackets elements were assembled in ground and then lifted via multiple cranes. The planning and execution of critical lifts involving multiple cranes required a careful and strategic approach to ensure optimal safety and precision. The team coordinated the intricate interplay of the various crane systems. This includes detailed job safety analysis for each operation, thorough scheduling, load distribution calculations, and real-time communication among the crane operators and the project team. By effectively orchestrating the synchronized movement of multiple cranes, the project successfully hoisted the heavy and complex loads, minimizing risks and potential disruptions while safeguarding the integrity of the structures being lifted. This intricate planning process underscored the Saudi Aramco's commitment to achieving operational excellence and safety in high-risk scenarios.

Figure 4—Jacket sub-structures fabricated at ground then lifted at site

Figure 5—Roll-over of main Jacket Row utilizing multiple cranes

Figure 6—Roll-over of main Jacket Row utilizing multiple cranes

Key Fabrication Achievements

Noteworthy achievements during the jacket fabrication process include:

- 50 legs placements in 43 days (8,100 metric tons).
- Fully painted jackets with total of 150,000 SQM of APCS 19C painting in 180 days.
- 63 safe and successful major critical lifts (8277 metric tons) in 65 days (25 grids numbers and 38 mud mats).
- 12 panels roll ups in 13 weeks with an average panel weight of 1,400 metric tons.

The fabrication journey ends with the successful load-out and sail-away of the Jackets. The load-out process of the jackets from the fabrication yard to the transportation barge is facilitated by Self-Propelled Modular Transporters (SPMTs), which represents a sophisticated operation characterized by meticulous coordination and engineering robustness. By utilizing SPMTs, the project achieved a highly controlled and maneuverable process, minimizing stresses on the jackets and ensuring their safe passage from fabrication to transportation.

Figure 7—Loadout of 2 of the 6 Jackets into the transportation barge

Conclusion and Recommendation

The fabrication of the Marjan GOSP-4 six major jackets represent a significant milestone in the Marjan Increment Program and in Saudi Aramco's offshore projects history. The unprecedented scale of the jackets combined with the stringent schedule requirements, and execution during a global pandemic, presented considerable challenges. Through the adoption of advanced construction methodologies, digital technologies, and the right welding techniques, these challenges were effectively mitigated, resulting in the timely completion of fabrication activities with zero lost time injuries. The application of automated welding processes, including the zero-root gap welding technique, delivered measurable gains in productivity and quality. While the QR based consumables management system provided real time control, accountability, and efficiency in materials management. Furthermore, the adaptation of augmented reality training for working at height coupled with the meticulous planning for the multi-crane lifting operations, reinforeced the project safety culture and enables the achievement of more than 9 million safe manhours. Collectively, the abovementioned not only enhanced project delivery but also established new performance benchmarks for offshore fabrication activities. Future offshore developments shall build on these outcomes by expanding the use of digital tracking systems, Augmented Reality application for safety training and enhanced welding techniques. Lessons learned from Marjan GOSP-4 provide a valuable reference for the planning and execution of upcoming Mega Offshore Projects.

Acknowledgment

Sincere appreciation to Mr. Khaled AlSulaim and Mr. Hamad AlShaiban for their invaluable support and expert guidance, which played a pivotal role in the successful design, fabrication, and construction of the Marjan GOSP-4 Project.

In addition, for clarity and conciseness, we acknowledge the use of ChatGPT, an AI tool as the text in some cases were re-written by the help of the tool for clarity and better structure. However, authors have done all required fact-checks to make sure the contents are very accurate and up the required meaning.